COGNITIVE·TERRAIN

BLUE vs OLVANA · THE WAR FOR HOW A NATION THINKS



NO SHOTS FIRED. SIXTEEN WEEKS TO WIN A MIND.

The complete rules, every screen explained, and the research behind it.

8 MOVES DECODED

Plant Stories to Lessons Learned: what each card does and where it hits hardest.

SWAP CODES, NOT SERVERS

Play a friend over email or chat. No accounts needed.

EDUCATIONAL WARGAME · TURN-BASED
WINDOWS · MAC · LINUX · ANY BROWSER

CT · 0013 · 2026

CONTENTS

INSIDE THIS ISSUE

WELCOME TO THE WAR ROOM

03 THE BIG IDEA
Cognitive warfare and the decision loop in one page

04 GETTING STARTED
Install, pick a scenario, choose a difficulty

05 ANATOMY OF THE SCREEN
Thirteen things on screen, numbered and explained

07 READING THE MAP
Dots, rings, health bars and the side panels

08 YOUR TURN, STEP BY STEP
Pick, aim, preview, strike

10 THE ARSENAL
All eight moves, with fact files

12 DEFENDING YOUR OWN BRAIN
Threats, warnings and what you lose

13 THE SPREAD SCORE
Views, likes, comments, shares and grades

14 NEW TECHNOLOGIES & CRISES
The four weeks that decide the campaign

16 TWO PLAYERS
Play a friend by swapping turn codes

18 SETTINGS & ACCESSIBILITY
Plain or military terms, colour-safe mode, pace

19 SCENARIOS
Four campaigns and their crises

20 KNOW YOUR ENEMY
How Blue and Olvana differ, plus tips & tactics

21 GLOSSARY, SOURCES & QUICK REFERENCE
Plain-language terms, credits, a one-page cheat sheet

FROM THE BRIEFING ROOM

Countries compete in many ways short of war. One of them is reaching into the other side's newsfeeds, group chats and briefing rooms to change what people notice, what they believe and how fast their leaders decide.

Cognitive Terrain turns that contest into a board game you can finish over a lunch break. The board is two brains drawn as star maps. Each dot is a real kind of person, group or institution: a newsroom, a cabinet, a crisis team, the public's trust in its own government.

You command **Blue**, an open democracy. Your opponent is **Olvana**, a closed, centralized state. Over sixteen weeks you will try to cloud Olvana's judgment while keeping your own clear. Four crises decide who comes out ahead.

Nobody fires a shot. Nobody needs to.

FIRST GAME?

Start with **Typhoon Ines** on **Recruit** and press **START WITH TUTORIAL**. It takes about ten minutes and points at everything on this spread.

PRIMER · READ THIS FIRST

HOW A COUNTRY MAKES UP ITS MIND

Every decision runs the same loop, from a town council facing a storm to a cabinet facing a crisis. Break the loop and you break the decision.



Picture a town hearing that a storm is coming. First people **notice** the warning. Then they **interpret** it: is it serious, who says so, have we seen this before? Leaders **decide** whether to evacuate, and finally people **act**. Under all of it sits **trust**. If nobody believes the weather office, every step goes worse.

Military planners call the four steps the **OODA loop**: Observe, Orient, Decide, Act. Researchers call the fight over that loop **cognitive warfare**, a long contest over how people perceive, interpret, decide and act. In this game each brain is split into those five areas, and every dot belongs to one of them.

Your score is not how many people saw your posts. It is how well each side makes decisions when it matters. That is the game's central lesson: **reach is not effect**.

THE FIVE AREAS, AND WHAT LOSING ONE COSTS YOU

When the other side holds half the dots in one of your areas, that area is **lost** and you suffer its penalty until you win it back.

AREA	WHAT IT IS IN REAL LIFE	IF YOU LOSE IT
NOTICE	Where people get information: news, social feeds, reports from the ground, intelligence briefings.	Too much noise: you can no longer see the other side's brain clearly.
INTERPRET	How people make sense of it: who is to blame, what it means, what it looks like from the past.	Endless arguing: every move costs 1 extra point.
DECIDE	Leaders and teams who weigh options and choose: cabinets, planners, appetite for risk, red lines.	Leaders stall: 1 fewer move per turn.
ACT	Carrying decisions out: troops, supplies, public cooperation, keeping promises.	Orders go out wrong: 1 in 4 of your moves fails.
TRUST	The slow foundation: trust in institutions, unity, shared identity, knowing where to check facts.	Official statements count half, recovery halves, and enemy posts spread further.

SETUP · FIVE MINUTES

BOOT UP & DEPLOY

INSTALL

SYSTEM	WHAT TO DO
WINDOWS	Run **CognitiveTerrain-0.13-Setup.exe**. If Windows says "protected your PC", choose *More info* → *Run anyway* (the playtest build is not code-signed yet). No admin rights needed.
MAC	Unzip the Apple Silicon or Intel build, drag the app to Applications, then right-click → *Open* the first time.
LINUX	Unpack the .tar.gz and run `./cognitive-terrain`.
ANY PC	Double-click the .html file. It opens in your browser. Nothing to install.

The game works offline and never needs an account. Settings, story history and saved two-player games stay in the browser or app you play in.

THE START SCREEN

1. **Scenario.** Four fictional campaigns. Typhoon Ines has no special rules, so it is the best first game. See page 19.
2. **Language. Plain** uses everyday words ("Decision Health", "Plant Stories"). **Military** uses doctrine terms ("DQ", "Seed Channels"). Switch any time.
3. **Difficulty. Recruit**: Olvana is slower and sloppier. **Standard**: a steady, sensible opponent. **Veteran**: sharper, richer, more aggressive.
4. **Start.** Pick the tutorial, skip straight to the game, read the in-game guide, or choose **TWO PLAYERS**.



FIG. 1 The start screen. Scenario, language, difficulty and sound are all set here before week 1.

ANATOMY OF THE WAR ROOM

EVERYTHING ON SCREEN, NUMBERED

The big brain in the middle is the one you are looking at; the small one, top right, is the other. Click either tab or the small brain to swap.



FIG. 2 Week 6 of a Typhoon Ines campaign, looking at Olvana's brain. Blue has lost ground in its own TRUST area (the warning in the top-right panel).

1 TOP BAR

Week, crisis points so far (Blue – Olvana), and the next crisis with the area it will test.

2 VIEW TABS

INFLUENCE OLVANA shows their brain. PROTECT BLUE shows yours; the ▲ count is how many of your dots are under pressure.

3 HEALTH BARS

Each side's Decision Health, 0 to 1, with marks at the crisis cut-offs. Label reads SOUND, DELAYED or ERROR.

4 BRAIN TITLE

Whose brain this is, its health, and a one-line hint about what you can do here right now.

5 THE BRAIN

Dots are people, groups and institutions; lines are who influences whom. Faint area names mark the five regions.

6 ENTRY POINT ◇

News Media, Social Media and Government Contacts. Every attack starts here or next to a dot you already lean.

7 TOP LEADERSHIP

The big core dot. Worth 2 budget a turn to whoever holds the other side's.

8 THE OTHER BRAIN

A live mini-map. On your own brain it lists the dots under pressure and suggests a defence.

9 WHAT IS HAPPENING

Health, decision speed, how close each side is to the upper hand, the five areas as bars, and the event log.

10 BUDGET & MOVES

Points to spend and moves left this turn. Hover a card to see what it would cost.

11 TIMELINE

The sixteen weeks. ◆ marks a new technology, CRISIS marks the four decision points.

12 ATTACK CARDS

Four moves that push the other brain. Keys 1 to 4.

13 DEFENCE CARDS

Four moves that protect your own brain. Keys 5 to 8.

14 END TURN

Pulses when you are out of moves or budget. In two-player games it hands you a turn code.

ANATOMY · CLOSE-UPS

THE INSTRUMENTS, UP CLOSE

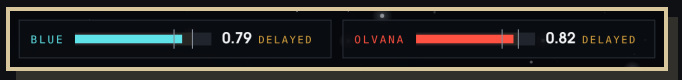


FIG. 3 Health bars. Blue on the left, Olvana on the right. The tick marks are the crisis cut-offs: above the right tick a crisis goes SOUND, between the ticks DELAYED, below them ERROR.



FIG. 4 View tabs and brain title. The "5 ▲" on PROTECT BLUE says five of your dots are being pushed. The hint line changes with the card you pick.



FIG. 5 Budget and moves. 8 of 18 points, 3 moves left, and +7 points arriving next turn. Budget above 18 is wasted, so spend it.



FIG. 6 The mini panel while you view Olvana. It watches your home brain for you: each ▲ row names a dot under pressure and the card that would help. The red-ruled warning means one more loss costs you an area.



FIG. 7 What is happening. Health and its label, how many hours a decision now takes, the race to the upper hand (LEAD 0/2), how much your side has learned, and how much of each area the brain still controls.

READING THE MAP

The brains are drawn like star charts on purpose: dots are nodes in a network, not parts of an anatomy lesson. Colour tells you who a dot leans toward; rings tell you what has been done to it.

SYMBOL KEY

- **Cyan dot.** Leans Blue. Brighter means stronger.
- **Red dot.** Leans Olvana.
- **Amber dotted ring.** Contested: nobody holds it.
- **Diamond ◇ with a white dashed ring.** An entry point that has been **prebunked**: incoming pushes count half.
- **Corner brackets.** Watched by a Watch Team: attacks land weaker, corrections harder.
- **Blinking amber.** Framed: it will not recover for a while.
- **Large dot.** Top Leadership, the core of the brain.
- **▲ on your brain.** Under pressure. Defend it soon.

HIDDEN BY DESIGN

Only entry points and leadership are named all the time. Hover any dot to see who it is. Settings can turn names on for every dot.



FIG. 8 Olvana's brain at week 6. Three entry points (◇) and Top Leadership are labelled. Social Media at the bottom is contested and prebunked; the faint words are the area names.

HOLDING, LEANING AND LOSING

Every dot has a control score from −100 (fully Olvana) to +100 (fully Blue). At **30** or more toward a side, that side **holds** it. At **10** or more you have a **foothold**, and your attacks can reach the dots linked to it. Hold half the dots in an area and the area is **lost** for its owner.

At the end of each week, dots drift back toward their own side, and strongly held dots push on their neighbours. A foothold you ignore will fade; a cluster you hold will spread.

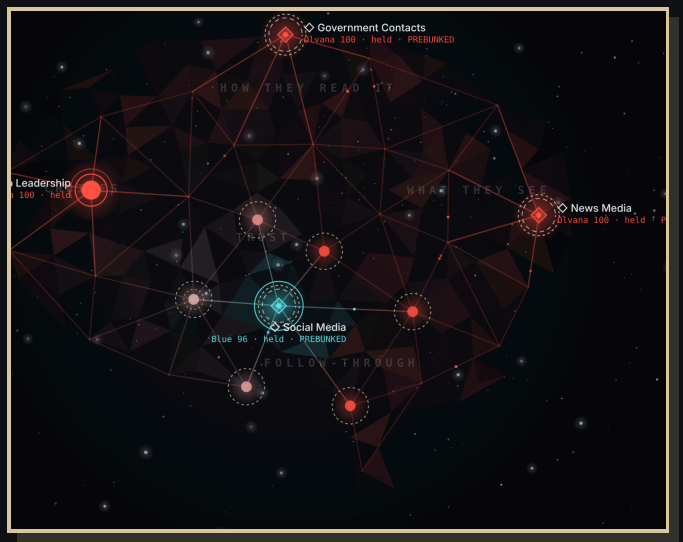
WALKTHROUGH · ONE MOVE FROM START TO FINISH

PICK. AIM. PREVIEW. STRIKE.

Each week you get three moves and a budget. Here is one move, start to finish.



STEP 1 · PICK A CARD Click a card in the bottom bar or press 1 to 8. Red-family attacks on the left, defences on the right. Each shows its cost in points. Picking an attack jumps the view to Olvana; a defence jumps it to your brain.



STEP 2 · FIND A RINGED DOT With Plant Stories selected, the dots you can reach this turn are ringed. Entry points are always in reach; other dots need a foothold next door.



STEP 3 · HOVER TO PREVIEW The tooltip says who the dot is, who holds it, what it does in real life, and why this move works here. Click to commit.



STEP 4 · SEE WHAT HAPPENED A result card shows the headline, the spread grade (here D), the audience it actually reached, and how far opinion moved. FULL STORY opens the complete report.

PRO TIP

Each card's best areas multiply its punch by up to 1.5. Plant Stories hits hardest in **NOTICE**, Frame the Story in **INTERPRET**, Bot Swarm in **ACT**.

MOVES CAN FAIL

About 1 in 12 attacks is taken down or falls flat, more often on watched or prebunked dots. A failed attack nudges the dot *toward* the defender. Official Statements are sometimes ignored, especially once your TRUST area is lost.

BUDGET · MOVES · THE WEEKLY TICK

SPEND IT OR LOSE IT



FIG. 9 Three moves, 8 points.



FIG. 10 Out of moves with points left over: the **+1 MOVE** button appears. Blue can buy one extra move a turn for 2 points.

DON'T HOARD

Your budget tops out at **18**. Anything past that is thrown away at the start of your next turn. If the bar is nearly full, surge.

WHERE POINTS COME FROM

SOURCE	PER TURN
Base income	+6
Each enemy dot you hold	+1
Their Top Leadership, if you hold it	+2
Their Government Contacts entry point	+2

Most cards cost 2. Bot Swarm and Lessons Learned cost 3. Lose your INTERPRET area and everything costs 1 more. Lose DECIDE and you get only 2 moves a turn.

THE WEEK, IN ORDER

1. **New technology** (weeks 2, 6, 10, 14): both sides choose to use it or prepare for it.
2. **Blue's turn.** Collect budget, make up to 3 moves (plus a surge), press END TURN.
3. **Olvana's turn.** Its moves pop up as small cards at the top of the screen while your brain is shown.
4. **Upkeep.** Every dot recovers a little toward its own side, unless framed. Strongly held dots push on their neighbours. Lessons Learned makes your recovery faster.
5. **Crisis** (weeks 4, 8, 12, 16): both sides are scored. See page 15.

WATCH THE CLOCK

The top bar always names the next crisis and the area it tests, e.g. `CRISIS 2 · 2 WK · INTERPRET`. That area is where you need to be strong in two weeks.

ATTACK CARDS · KEYS 1–4

FOUR WAYS TO GET INSIDE

Attacks go on the other side's brain. Each is a real family of influence technique drawn from the research behind the game.

PLANT STORIES

KEY 1

GET THE MESSAGE IN · MILITARY: SEED CHANNELS

COST	PUSH	TARGETS	SPECIAL
2	20	1	×1.5 ◇

NOTICE ×1.5 | INTERPRET | DECIDE ×.75 | ACT ×.75 | TRUST

Place content in the news and feeds people already use. The strongest single push in the game, and 50% stronger on an entry point. Your opening move.

> IN REAL LIFE: look-alike local news sites, sponsored posts, paid influencers.

FRAME THE STORY

KEY 2

SHAPE THE STORY · MILITARY: FRAME & PRIME

COST	PUSH	TARGETS	SPECIAL
2	16	1	FRAMED

NOTICE ×.75 | **INTERPRET ×1.5** | **DECIDE ×1.25** | ACT | **TRUST ×1.25**

Tell it in a way that triggers emotion or identity. The target stops recovering for 2 turns, so a follow-up hit sticks. The setup move before a capture.

> IN REAL LIFE: a clip edited and captioned to make people angry or afraid.

BOT SWARM

KEY 3

MAKE IT LOUD · MILITARY: COORDINATED AMPLIFICATION

COST	PUSH	TARGETS	SPLASH
3	12	+LINKS	7 EACH

NOTICE | INTERPRET | DECIDE | **ACT ×1.5** | **TRUST ×1.25**

Many coordinated accounts repeat it at once. Hits the target *and every dot linked to it*. Hold their Social Media entry point and the splash hits at full strength.

> IN REAL LIFE: thousands of accounts posting the same thing within an hour.

REPOST & REBRAND

KEY 4

ADAPT · MILITARY: PIVOT & RESEED

COST	PUSH	TARGETS	SPECIAL
2	10	+LINKS	STRIP

NOTICE | INTERPRET | DECIDE | ACT | TRUST

Come back with new accounts and a new angle after removals. Strips **Watched** and **Prebunked** from the target and its links. Use it when the other side digs in.

> IN REAL LIFE: a banned network reappearing under new names the next day.

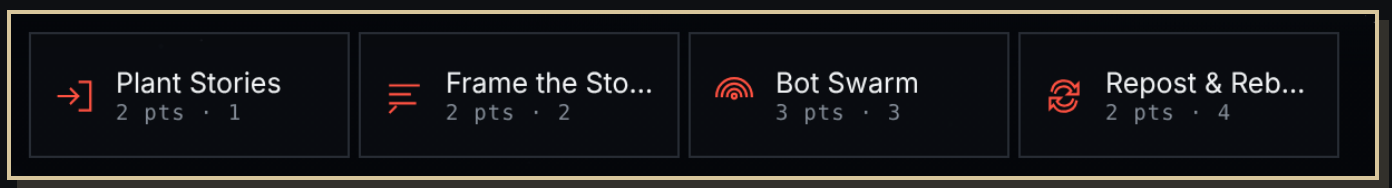


FIG. A The attack half of the card bar. Each card shows its icon, cost and hotkey; a card you cannot afford is dimmed.

COMBO

Frame the Story a dot this week, then **Plant Stories** on it next week: it cannot recover in between, so the second push usually captures it. If it is prebunked, open with **Repost & Rebrand**.

DEFENCE CARDS · KEYS 5–8

FOUR WAYS TO HOLD THE LINE

Defences go on your own brain. Attacking is cheap and defending is not: the research calls it the attacker's cost advantage. Spend where the next crisis will look.

WATCH TEAM

KEY 5

SPOT IT · MILITARY: ASSESSMENT CELL

COST	LASTS	TARGETS	BOOST
2	3 WK	+LINKS	×1.5

NOTICE ×1.5 · INTERPRET · DECIDE · ACT · TRUST

Watches a dot and its links. Incoming posts land a grade weaker and are more likely to be exposed, framing is cleared, and your Official Statements there hit 50% harder.

> IN REAL LIFE: analysts tracking sudden spikes of identical posts.

PREBUNK

KEY 6

BUILD IMMUNITY · MILITARY: PREBUNK & INOCULATE

COST	LASTS	TARGETS	INCOMING
2	3 WK	+LINKS	×0.5

NOTICE · **INTERPRET ×1.5** · DECIDE · ACT · TRUST

Teach people the trick before they see it. A whole cluster takes half influence for three weeks. Best laid down *before* the pressure arrives.

> IN REAL LIFE: short videos showing how fake clips are made, shown before an election.

OFFICIAL STATEMENT

KEY 7

SET THE RECORD STRAIGHT · MILITARY: AUTHORITATIVE RESPONSE

COST	RESTORE	TARGETS	IF WATCHED
2	14+	1	×1.5

NOTICE · INTERPRET · **DECIDE ×1.5** · ACT · TRUST

Correct the record through trusted voices. Pulls one dot back toward you. Blue's free press makes it hit extra hard while Blue's TRUST holds, and it counts half once TRUST is lost.

> IN REAL LIFE: a press conference with evidence from a trusted spokesperson.

LESSONS LEARNED

KEY 8

LEARN & PRACTICE · MILITARY: AFTER-ACTION REVIEW

COST	ALL DOTS	TARGET	MAX
3	+4	NONE	3×

NOTICE · INTERPRET · DECIDE · **ACT ×1.5** · TRUST

Review what went wrong and drill. Every dot you own gains 4, framing is cleared, and your weekly recovery gets permanently faster (up to three times). Pays more the earlier you buy it.

> IN REAL LIFE: an after-action review followed by a practice exercise.

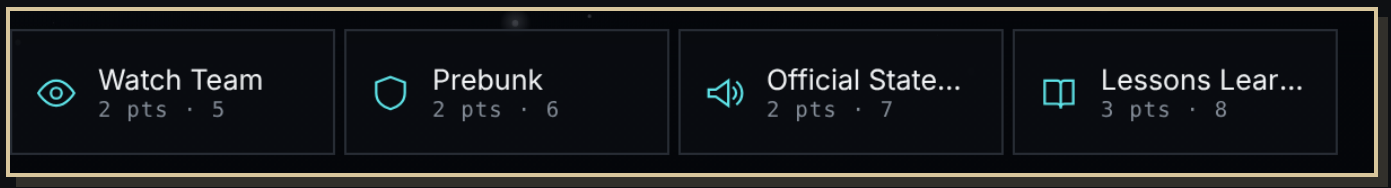


FIG. B The defence half of the card bar. Lessons Learned needs no target: click it and it fires.

PRO TIP

Watch Team first, Official Statement second, on the same dot in the same turn: the statement lands 50% harder. Against a Bot Swarm wave, Prebunk the cluster instead of fixing dots one by one.

PROTECT YOUR OWN BRAIN

Olvana never skips a turn. If you only attack, your own Decision Health will slide by week six and the crises will punish you for it.

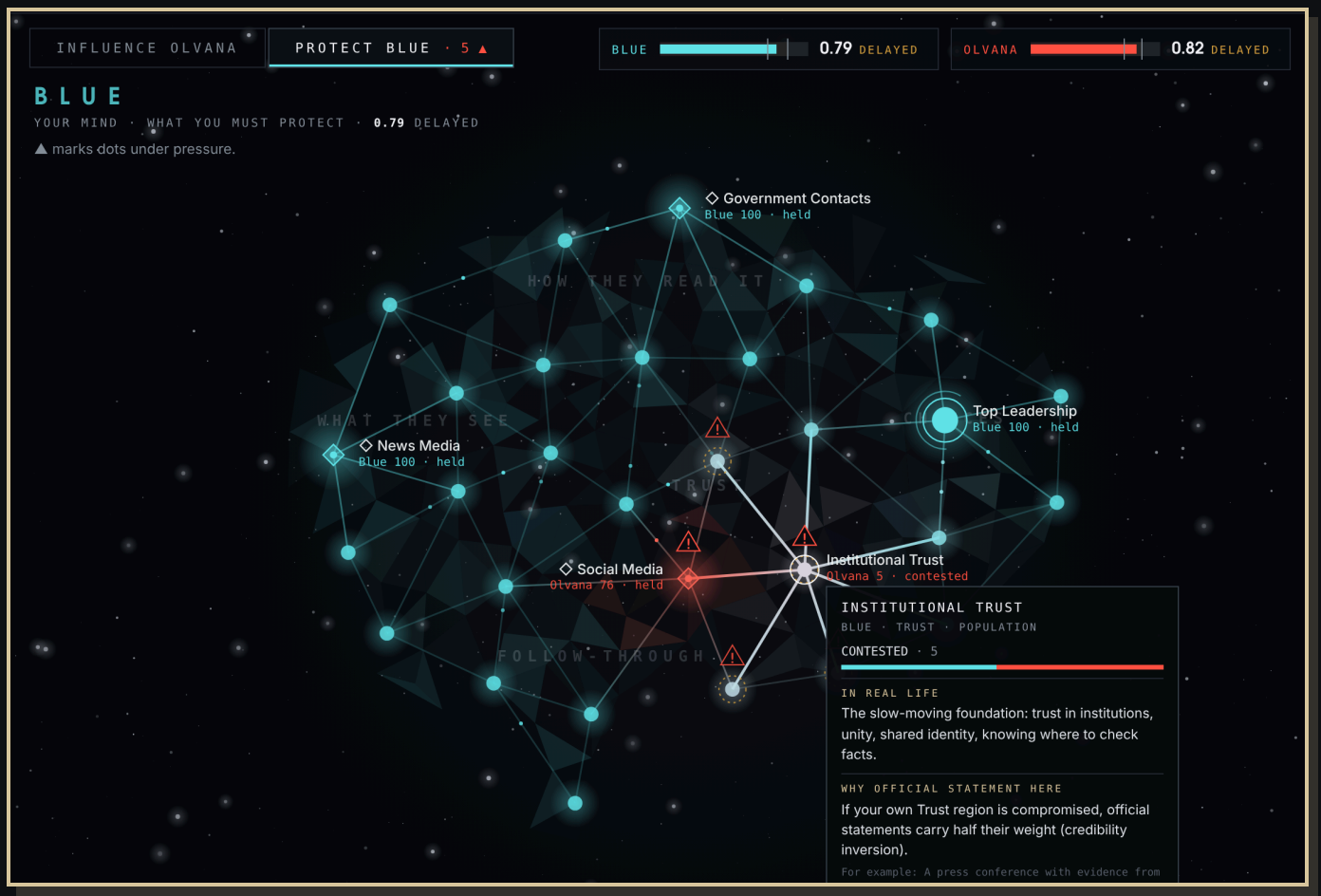


FIG. 11 PROTECT BLUE with Official Statement selected. ▲ marks dots under pressure; the tooltip previews the effect on Institutional Trust and explains in real-life terms why a statement helps there.

Switch to **PROTECT BLUE** at least once a turn, or simply read the mini panel while you attack: it lists the three dots in most trouble and suggests a card for each. The tab shows the count, like `PROTECT BLUE · 5 ▲`.

When an area is one dot from falling, the mini panel shows a red-ruled warning naming the penalty you are about to take. That is the moment to spend, even if it means skipping an attack.

Think about **where the next crisis looks**. A crisis scores 60% on your overall Decision Health and 40% on the one area it tests. Two weeks before a crisis, put your defence there.

Of all your areas, **TRUST** is the one to guard. It falls slowly but costs the most: your statements count half, your recovery halves, and every Olvana post spreads a grade further. Blue's free press helps, because TRUST recovers faster for Blue than any other area.

THE NUMBERS BEHIND EVERY POST

VIEWS ARE CHEAP. SHARES COUNT.

Every attack is a small campaign of posts. The game rolls the audience it reached, then grades the campaign the way the research does.

SPREAD SCORE PER POST

$$(0.1 \cdot \text{views} + 0.3 \cdot \text{likes} + 0.7 \cdot \text{comments} + 1 \cdot \text{shares}) \div \text{posts}$$

RUSHING & XU (2025) · ENGAGEMENT EFFECTIVENESS

GRADE	SCORE	MEANING	EFFECT
A+	10,000+	Viral	×1.6
A	1,000+	Excellent	×1.4
B	100+	Above average	×1.2
C	10+	Average	×1.0
D	4–9	Below average	×0.7
E	3	Poor	×0.45
F	0–2	Failure	×0.25

Each move type has its own crowd. **Plant Stories** draws lots of views and few shares. **Frame the Story** sparks the most comments. **Bot Swarm** racks up shares but little real conversation. Better posts earn more interaction per view, so a viral post looks viral in the numbers too.

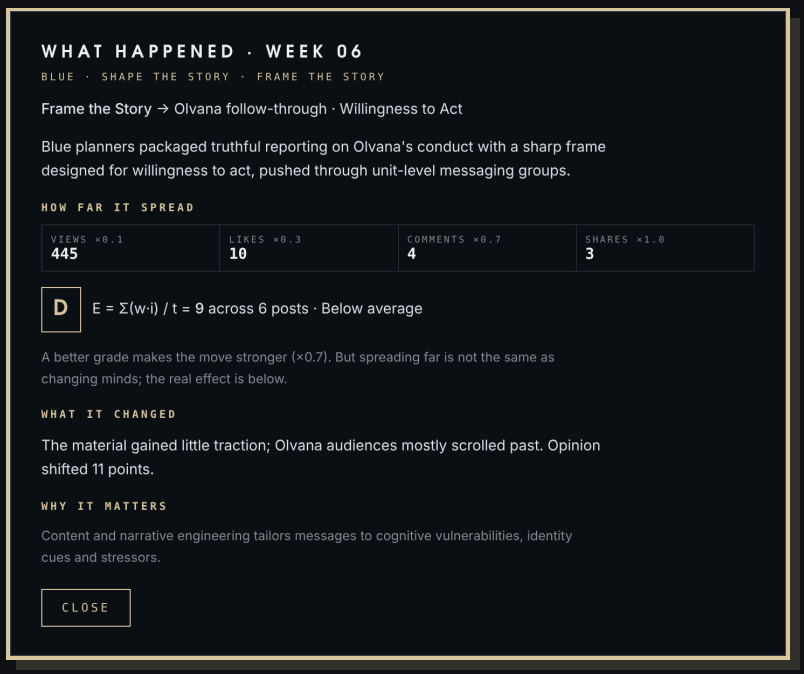


FIG. 12 The full story (◉ in the log, or FULL STORY on a result card): audience counts with their weights, the grade and its formula, what changed and why it matters.

WHAT PUSHES THE GRADE

UP	DOWN
Posting on an entry point	A Watch Team on the target (−1 grade)
Holding their News Media	Olvana's censored internet (Blue's posts)
Their TRUST area lost (+1 grade)	
A new technology you chose to use	

REACH ≠ EFFECT

A grade only multiplies the push. A viral post on the wrong area still loses the crisis. Aim first, then worry about virality.

WEEKS 2 · 6 · 10 · 14

NEW TECHNOLOGY ARRIVES



FIG. 13 The timeline. ◆ marks a new technology; CRISIS 1–4 fall in weeks 4, 8, 12 and 16, and crisis 4 counts double (×2).

Four times a campaign a new technology changes the board. Both sides choose in secret: **USE IT** for an attack bonus, or **PREPARE DEFENSES** to deny the other side's bonus and prebunk your three entry points for two weeks.

If you use it and they prepared, you get nothing. If you prepared and they used it, they get nothing against you. The choices show in the log, and in the New Technologies list in the side panel.

TECHNOLOGY	BOOSTS
ALGORITHMIC TARGETING	Frame the Story +1 grade
SOCIAL BOTS	Bot Swarm +1 grade, full splash
TROLL FACTORIES	Bot Swarm, Repost & Rebrand +1 grade
AI SYNTHETIC MEDIA	Every attack +1 grade
HUMANOID ROBOTS	Plant Stories reaches any NOTICE or INTERPRET dot, no foothold needed

Each campaign uses four of these. The attribute scores come from Rushing, Umeokolo & Xu (2025), whose forecast predicts a new disruptive innovation roughly every 4.6 years.

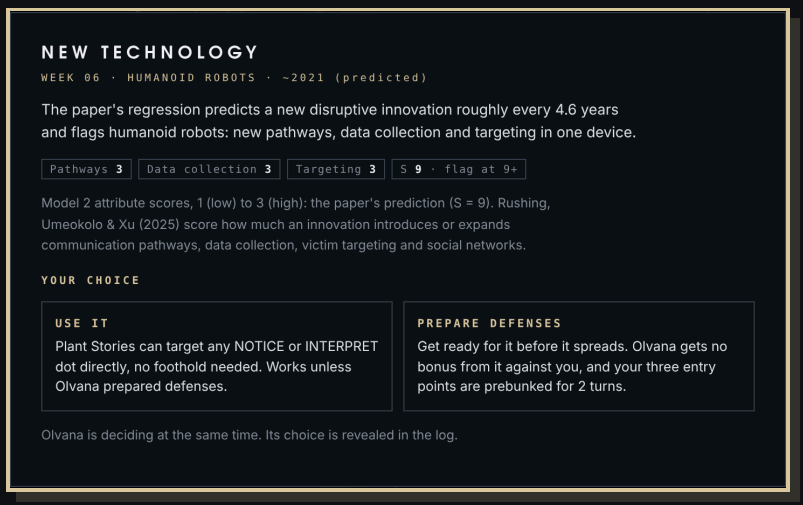


FIG. 14 Humanoid Robots arrive in week 14. The attribute scores (pathways, data collection, targeting) sum to S = 9, the paper's threshold for a flagged innovation.

WEEKS 4 · 8 · 12 · 16 · HOW TO WIN

FOUR MOMENTS DECIDE IT ALL

At the end of weeks 4, 8, 12 and 16 a crisis hits the scenario's setting: a contaminated-water rumour, a severed cable, a fake evacuation alert. Each crisis tests one area, named in the top bar weeks ahead.

Each side's crisis result comes from **60% overall Decision Health + 40% of the tested area**, with an extra penalty if that area is lost.

RESULT	NEEDS	POINTS
SOUND	0.86 or better	3
DELAYED	0.72 – 0.85	1
ERROR	below 0.72	0, and the other side gets 1

Crisis 4 counts **double**. Most points after week 16 wins; a tie goes to the healthier side.

THE KNOCKOUT

Keep the other side at least **two areas worse off** than you for **two weeks running** and you win on the spot. The side panel tracks it as LEAD 0/2.

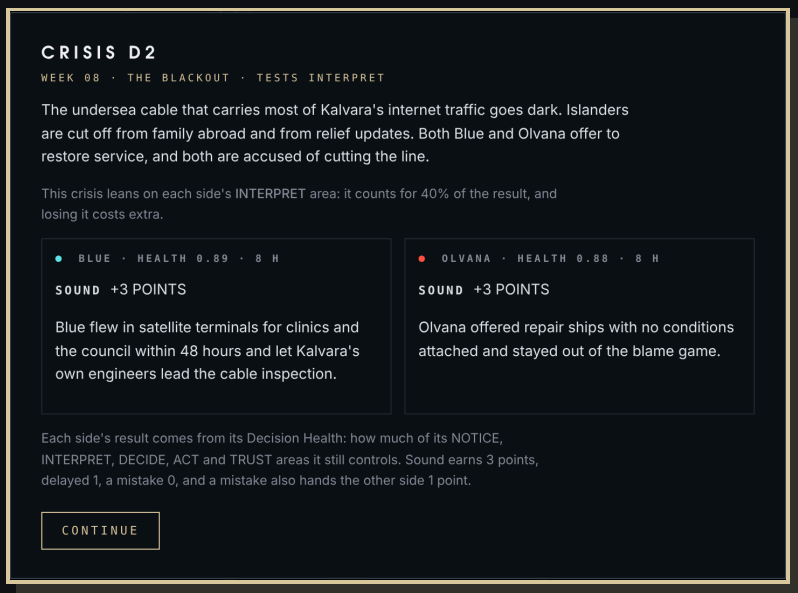


FIG. 15 Crisis 2, The Blackout. Both sides handled it soundly and took 3 points each. The text says what each side actually did.

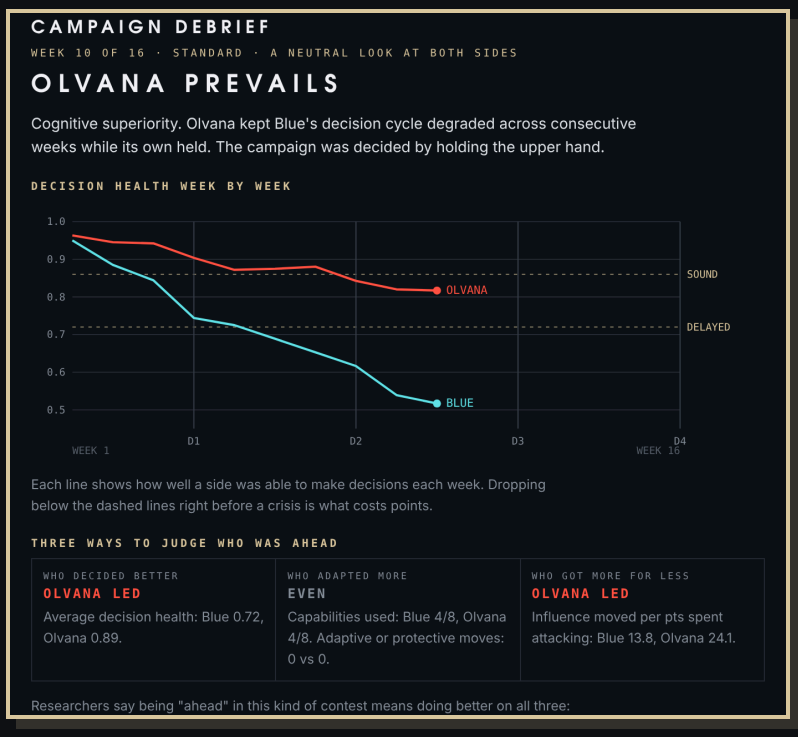


FIG. 16 The debrief: a neutral look at both sides, health week by week, three ways to judge who led, the turning points and lessons for next time.

HEAD TO HEAD · NO SERVER REQUIRED

PLAY A FRIEND BY SWAPPING CODES

NEW IN 0.13

Each turn ends with a short line of text. Send it any way you like: email, chat, Teams. Your opponent pastes it in, watches your moves play out, takes their turn and sends one back.

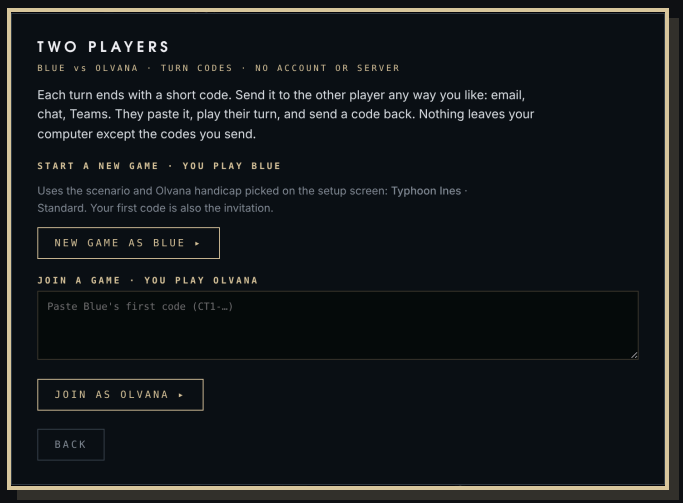


FIG. 17 TWO PLAYERS on the start screen. Blue starts a game; Olvana joins by pasting Blue's first code. Saved games appear under RESUME.

BLUE STARTS

1. Pick a scenario and Olvana handicap on the setup screen.
2. **TWO PLAYERS → NEW GAME AS BLUE.**
3. Play your moves and press **END TURN · GET CODE**.
4. **COPY CODE** and send it. This first code is also the invitation.

OLVANA JOINS

1. **TWO PLAYERS**, paste Blue's code, **JOIN AS OLVANA**.
2. Blue's moves replay on your screen, then it is your turn.
3. End your turn, copy your code, send it back.

THEN, EVERY WEEK

Paste their code, play, send yours. Repeat until week 16. When Olvana's last turn ends the game, Olvana gets one final code to send so Blue's game ends too.

SEND. PASTE. REPEAT.

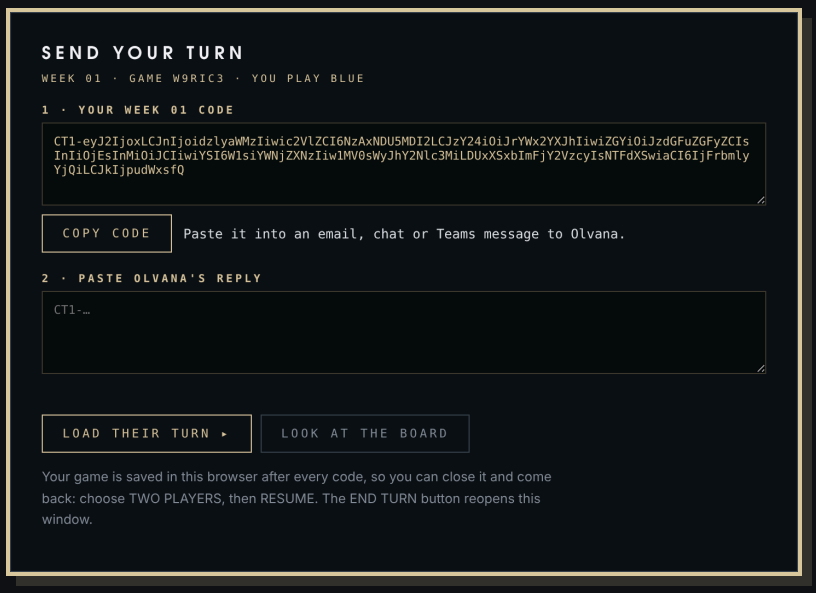


FIG. 18 After END TURN: your code at the top with a COPY button, and a box for your opponent's reply below. LOOK AT THE BOARD closes the window; the END TURN button (now TURN CODES ▸) brings it back.

Q&A

WHAT IS INSIDE A CODE?

Your moves, your new-technology choice, the game's ID and starting seed, and a fingerprint of the board. Nothing personal and nothing from your computer.

DO WE NEED THE INTERNET?

Only to send the code. The game itself runs offline.

CAN WE USE DIFFERENT VERSIONS?

You can mix the Windows app, the Mac app and the HTML file, as long as both of you are on build 0.13.

I CLOSED THE GAME MID-MATCH.

Your game saves after every code. Open **TWO PLAYERS** and pick it under **RESUME**.

"BOARD CHECK FAILED" IN THE LOG?

The two copies disagree. Compare health bars with your opponent; if they differ, restart from the last code you both agree on.

HOUSE RULES

Neither of you sees the other's moves until the code arrives, and new-technology choices stay secret until both are in. That makes it a good fit for a seminar: pair people up, give them a week, and run the debrief together at the end.

WHAT'S DIFFERENT FROM SINGLE PLAYER

- Whoever plays Olvana sees the game from Olvana's side: their brain is "yours", the tabs say PROTECT OLVANA and INFLUENCE BLUE.
- Your own moves get the big result cards; your opponent's appear as small cards at the top of the screen.
- The tutorial is single-player only. Your two-player game stays saved while you take it.

⚙ IN THE TOP BAR · ACCESSIBILITY

MAKE IT YOURS

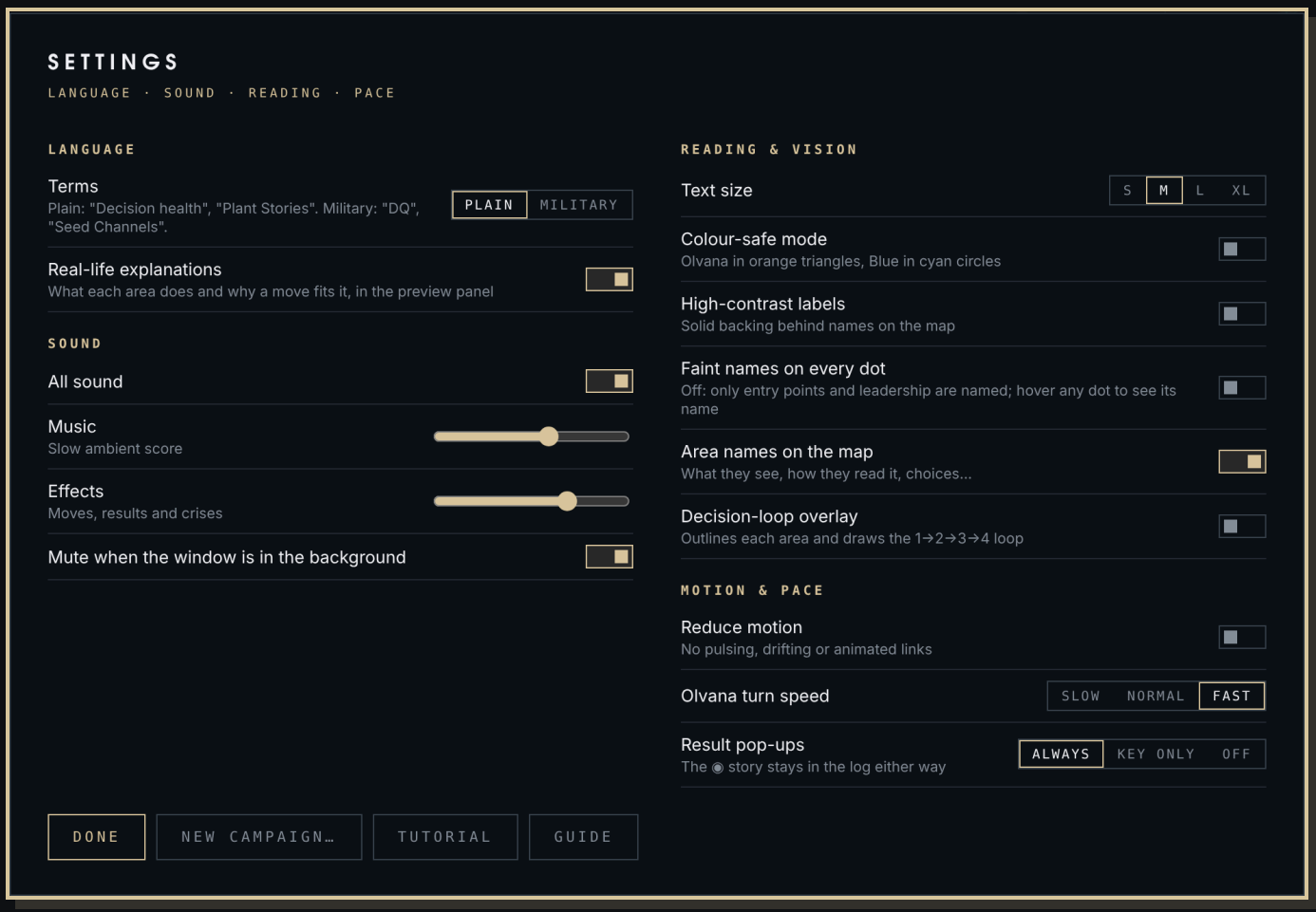


FIG. 19 Settings: language, real-life explanations, sound, text size, colour-safe mode, high contrast, map labels, the decision-loop overlay, reduced motion, Olvana's turn speed and how often result cards pop up.



FIG. 20 Colour-safe mode: Olvana becomes orange triangles, Blue stays cyan circles, so the sides differ by shape as well as colour.

PLAIN OR MILITARY?

PLAIN	MILITARY
Decision Health	Decision Quality (DQ)
Budget · pts	Influence points · IP
Notice / Interpret	Observe / Orient
Entry point	Gateway
Plant Stories	Seed Channels
Watch Team	Assessment Cell
Crisis	Decision point
What happened	SITREP

In military mode Olvana's moves use its own doctrine names ("Public Opinion Warfare", "Network Swarming").

QUIET PLEASE

The music and sound buttons sit in the top bar. Mute when in background is on by default.

FOUR CAMPAIGNS · ALL ORIGINAL FICTION

CHOOSE YOUR BATTLEGROUND

Same rules, different pressure. Each scenario has its own crises, its own tested areas and its own starting conditions.

TYPHOON INES

DISASTER RELIEF · THE KALVARA ISLANDS

A typhoon flattens a small island republic that trades with both sides. Over sixteen weeks of relief and rebuilding, Kalvara chooses whom to trust.

> TWIST: none. The best first campaign.
> CRISES: The Relief Flotilla (NOTICE) · The Blackout (INTERPRET) · The False Alarm (ACT) · The Reconstruction Compact (TRUST, ×2)

THE VALMORA VOTE

A SNAP ELECTION · VALMORA

A partner democracy holds a snap election. Olvana quietly favours the candidate who would cancel the Blue trade pact.

> TWIST: both TRUST areas start weakened (Blue's more), and AI Synthetic Media arrives first.
> CRISES: The Leaked Audio (INTERPRET) · The Ballot Rumor (NOTICE) · Count Night (DECIDE) · Transfer of Power (TRUST, ×2)

COLLISION AT VARRIN REEF

MARITIME INCIDENT · TELUGA

An Olvanan cutter and a fishing boat from Teluga, a small Blue ally, collide at a disputed reef. Two fishermen are missing.

> TWIST: both sides' Government Contacts start weakened, so the fight over envoys starts on day one.
> CRISES: Two Videos (NOTICE) · The Missing Crew (ACT) · The Drills (ACT) · The Tribunal Ruling (DECIDE, ×2)

FEVER IN PORT SABRE

AN OUTBREAK · PORT SABRE

A fast-spreading virus hits a trading city-state that depends on both sides. Rumours move faster than the virus.

> TWIST: infodemic. Held dots push harder on their neighbours and both NOTICE areas start strained.
> CRISES: Patient Zero (INTERPRET) · The Bad Batch (TRUST) · The Forged Order (ACT) · The Health Accord (DECIDE, ×2)



FIG. 21 The scenario picker on the start screen. The highlighted box is the one you will play.

REPLAY VALUE

Every campaign draws a fresh brain map, and the headlines come from a large story bank written for each scenario, so you can play many games before stories repeat.

TRY THIS

Play the same scenario on **Standard** twice: once attacking every turn, once defending first. Compare the two debriefs.

ASYMMETRIC WARFARE

KNOW YOUR ENEMY

Blue and Olvana are not mirror images. Olvana is modelled as a closed, centralized one-party state; Blue as an open, rule-bound democracy.

OLVANA

CENSORED INTERNET	Blue's posts spread less; Olvana's entry points recover 20% faster.
STATE-FUNDED CONTENT	Bot Swarm costs 1 less. On Veteran, +1 budget a turn.
ALREADY INSIDE	Blue's Social Media entry point starts weakened (80, not 100).
RUN FROM THE TOP	Olvana's health leans 40% on DECIDE (Blue's: 30%). Fast when intact, brittle when shaken.
BAD NEWS TRAVELS SLOWLY	Lessons Learned heals +2 instead of +4 and builds recovery half as fast.

BLUE

OPEN INTERNET	Olvana's posts into Blue spread further.
FREE PRESS & COURTS	Blue's TRUST area recovers faster, and Official Statements hit harder while it holds.
THE TRUTH STICKS	Olvana dots won over to Blue drift back at about a third of normal speed.
CHEAP SURGE	Blue's extra move costs 2 points; Olvana's costs 5.

TIPS & TACTICS FROM THE WAR ROOM

1. **Open on an entry point.** Plant Stories on their News Media or Social Media gives you the foothold every later move needs.
2. **Hit DECIDE.** Olvana's centralized command makes it the biggest lever on its health.
3. **Read the top bar.** The next crisis names its area. Defend that area in the two weeks before.
4. **Frame, then plant.** Framed dots cannot recover, so the second hit captures.
5. **Buy Lessons Learned early.** Faster recovery compounds every week.
6. **Never sit on 18 points.** Surge instead.
7. **Prebunk before, not after.** Half damage is worth more on a cluster that is about to be hit.
8. **Guard TRUST.** It is slow to fall and painful to lose.

JARGON BUSTER

SAY IT IN PLAIN WORDS

TERM	PLAIN MEANING	IN THE GAME
INFORMATION OPERATIONS	Using information to affect what an opponent knows, believes and decides, while protecting your own side.	The whole game
COGNITIVE WARFARE	A long contest over how people perceive, interpret, decide and act.	The whole game
OODA LOOP	Observe, Orient, Decide, Act: the four steps of any decision.	NOTICE, INTERPRET, DECIDE, ACT
MISO / PSYOP	Military Information Support Operations: truthful messages aimed at foreign audiences.	Plant Stories, Frame the Story
PUBLIC AFFAIRS	Official communication with the public and press.	Official Statement
OPSEC	Protecting information an opponent could exploit.	Watch Team
PREBUNKING	Teaching people a manipulation trick before they meet it.	Prebunk
AMPLIFICATION	Making a message louder: bots, troll networks, coordinated sharing.	Bot Swarm
SYNTHETIC MEDIA	AI-generated text, images, audio or video.	New technology events
FRAMING	The angle a story is told from.	Frame the Story
ATTRIBUTION	Deciding, and saying publicly, who was responsible.	INTERPRET area
ENGAGEMENT EFFECTIVENESS	A research score for how far a post spread: shares count most, then comments, likes, views.	Spread grade F to A+
COGNITIVE SUPERIORITY	Staying ahead over time: deciding well, adapting faster, getting results cheaper.	LEAD; the knockout win
ACUTE vs CHRONIC	Fast effects (hours to days) versus slow ones (months to years).	Most areas vs TRUST
GATEWAY	A channel into a society's information space.	Entry points ◇
SITREP	Situation report.	◉ in the log
THREE WARFARES	Chinese military concept: public opinion, psychological and legal warfare.	Olvana's card names (military mode)
ODIN / DATE / OPFOR	US Army training environment and the fictional opposing forces it defines, including Olvana.	Olvana

SOURCES · ATTRIBUTION · LICENSE

WHERE THIS COMES FROM

The game is built on published research. The rules simplify it; the debrief and the in-game guide point back to it.

REFERENCES

- Rushing, B., Hersch, W., & Xu, S. (2026). *Cognitive Warfare: Definition, Framework, and Case Study.* arXiv:2603.05222.
- Rushing, B., & Xu, S. (2025). *Quantifying the Engagement Effectiveness of Cyber Cognitive Attacks: A Behavioral Metric for Disinformation Campaigns.* arXiv:2510.15805.
- Rushing, B., Umeokolo, M.-R., & Xu, S. (2025). *Towards Proactive Defense Against Cyber Cognitive Attacks.* arXiv:2510.15801.
- Headquarters, Department of the Army (2018). *ATP 3-13.1, The Conduct of Information Operations.*
- NATO Science & Technology Organization. *Cognitive Warfare: NATO Chief Scientist Research Report.*
- Shin, D. W. (2026). *Viewing China's Cognitive Warfare Through the Lens of Its Global Strategy.* National Intelligence University.
- Eads, L. J., Clarke, R., Lin, X. S., & McCreight, R. (2023). *Warfare in the Cognitive Age: NeuroStrike and the PLA's Advanced Psychological Weapons & Tactics.*
- Roozenbeek, J., & van der Linden, S. (2020). *Breaking Harmony Square: A game that inoculates against political misinformation* (cited in Rushing et al.).

ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

Changes made: the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; three of the five innovation score sets are the designer's own estimates; all scenario, narrative and tutorial text is original.

No endorsement: the authors and their institutions have not reviewed or endorsed this game.

Olvana is the opposing force from the US Army's ODIN DATE training environment. All four scenarios, their places (the Kalvara Islands, Valmora, Teluga and Varrin Reef, Port Sabre) and every event in them are original fiction. Any resemblance to real places or events is coincidental. Game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.

CUT OUT AND KEEP

QUICK REFERENCE CARD

EACH TURN

- 3 moves (2 if DECIDE is lost)
- +6 budget, +1 per enemy dot held
- Budget caps at 18
- Surge: +1 move for 2 pts (Blue)
- Keys 1–8 pick cards; Esc cancels

CONTROL

- Hold a dot at 30+
- Foothold at 10+: reach its links
- Entry points ◇ always in reach
- Half an area held = area lost
- Dots recover a little each week

CALENDAR

- New technology: weeks 2, 6, 10, 14
- Crises: weeks 4, 8, 12, 16 (16 counts ×2)
- SOUND 3 · DELAYED 1 · ERROR 0
- Knockout: 2 areas ahead, 2 weeks running

CARD	KEY	COST	DOES	BEST IN
PLANT STORIES	1	2	Push 20; ×1.5 on entry points	NOTICE
FRAME THE STORY	2	2	Push 16; no recovery for 2 turns	INTERPRET, DECIDE, TRUST
BOT SWARM	3	3	Push 12 + 7 on every link	ACT, TRUST
REPOST & REBRAND	4	2	Push 10; strips Watched and Prebunked	Anywhere
WATCH TEAM	5	2	3 weeks: attacks weaker, statements ×1.5	NOTICE
PREBUNK	6	2	3 weeks: half influence on dot + links	INTERPRET
OFFICIAL STATEMENT	7	2	Restore 14+; half if TRUST is lost	DECIDE
LESSONS LEARNED	8	3	All your dots +4; faster recovery (max 3)	ACT

LOSE THIS AREA…	…AND THIS HAPPENS
NOTICE	You can no longer see the other brain clearly
INTERPRET	Every move costs +1
DECIDE	−1 move per turn
ACT	1 in 4 moves fails
TRUST	Statements half, recovery half, enemy posts +1 grade


Entry point


Prebunked


Contested


Your dot under pressure